

Lecture 7: Time Series
AR, MA, ARMA and NVAR
MLP, RNN and LSTM
Neural ODEs and Physics-Informed Networks

Deep Learning for Scientific Discovery



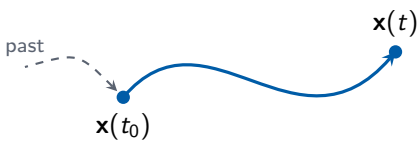
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Forecasting the Future

Laplace's deterministic ideal, 1814

"An intellect which at a certain moment would know all forces that set nature in motion, and all positions of all items of which nature is composed, if this intellect were also vast enough to submit these data to analysis, . . . for such an intellect nothing would be uncertain and the future just like the past would be present before its eyes."



- know all positions and velocities, $\mathbf{x}(t_0)$, and all forces, $\mathbf{F}$
- Newton's equations fix one future and one past
- classical physics is **deterministic**

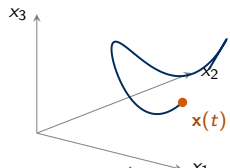
P.-S. Laplace, *Essai philosophique sur les probabilités* (1814)

State of a System

State and equations of motion

$$\mathbf{x}(t) = (x_1(t), \ldots, x_d(t)), \qquad \dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}), \quad \text{i.e. } \dot{x}_i = F_i(x_1, \ldots, x_d)$$

x_i: variables that change in time, together they fix the state; $\mathbf{F}$: deterministic, time-independent, smooth, in general nonlinear



state $\mathbf{x}(t) \in \mathbb{R}^d$, a point moving in phase space

- pendulum: angle, angular velocity
- weather: temperature, pressure, wind at every grid point
- laser: field, polarisation, inversion
- parameters (viscosity, heating) are constants inside $\mathbf{F}$
- components interact through $\mathbf{F}$; $\mathbf{x}(t)$ moves along a trajectory in phase space

Classical Mechanics: Newton's Equations as a Flow

From Newton to a first-order system

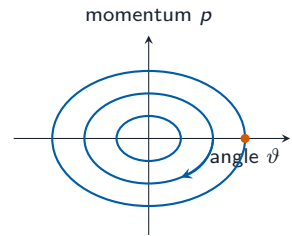
$$m\,\ddot{q} = f(q) \quad \Longrightarrow \quad \dot{q} = p/m, \quad \dot{p} = f(q)$$

$$\mathbf{x} = (q, p), \qquad \dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$$

q: position; p: momentum; f: force; the state $\mathbf{x}$ = all positions and momenta

Existence and uniqueness

for smooth $\mathbf{F}$ exactly one trajectory passes through every state



pendulum, small swings: each initial state lies on one closed orbit

- one degree of freedom gives two state variables; N particles in 3D give $6N$

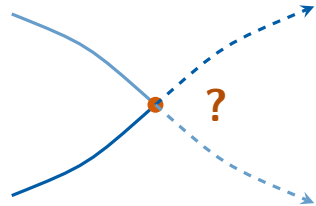
Trajectories Never Cross

full state space: allowed



one trajectory through each state

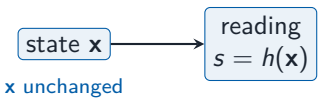
forbidden: a crossing



two futures from one state

- a crossing makes the **future ambiguous**: one state, two ways forward
- smooth **F** forbids it: in the full state space the future is unique
- in a measured projection curves do cross: one value does not fix the future, so we need past values (delay embedding)

Classical Measurement

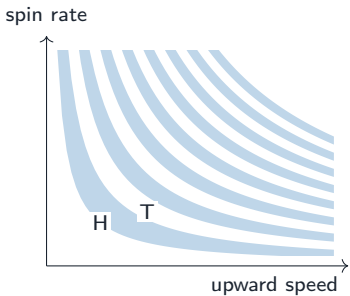


- an ideal measurement reads the state without changing it
- repeating it gives the same value
- better instruments give more digits, without limit in principle

Consequence

the state exists before we look, and every measured value is a function of it: with enough precision, the future is fixed

Classical Randomness: Missing Information



coin toss: initial conditions giving heads (white) and tails (blue), Keller

Epistemic randomness

the outcome is fixed by the initial conditions, but we do not know them precisely enough, so we describe it by a probability

- coin: speed and spin decide; a tiny error flips the result, so $p = 1/2$
- more information, better prediction: starting side up in about 51% of tosses
- probability measures our ignorance

J. B. Keller, Am. Math. Mon. 93, 191 (1986); P. Diaconis, S. Holmes, R. Montgomery, SIAM Rev. 49, 211 (2007)

Epistemic Randomness in Science

Epistemic: randomness of our knowledge, not of nature

the outcome is fixed by deterministic laws, but we lack the information to compute it; the probability changes when we learn more

unknown initial state
coin, dice: tiny changes of speed and spin flip the result

unresolved degrees of freedom
a pollen grain kicked by water molecules we do not track: Brownian motion, statistical mechanics

chaos
weather: tiny missing information grows, forecasts are run as ensembles from perturbed initial states

- test: more information, better prediction (quantum randomness fails it)
- in measured data, noise stands for everything we do not measure

The Quantum Realm

deterministic evolution, truly probabilistic measurement

configuration	two classical coins	two qubits: basis state	as a vector
00	both show 0	$\|0\rangle\|0\rangle = \|00\rangle$	$(1,0,0,0)^\top$
01	first 0, second 1	$\|01\rangle$	$(0,1,0,0)^\top$
10	first 1, second 0	$\|10\rangle$	$(0,0,1,0)^\top$
11	both show 1	$\|11\rangle$	$(0,0,0,1)^\top$

Coins: one configuration

the coins are in one of the four;
unknown which: probabilities $(p_{00}, p_{01}, p_{10}, p_{11})$
looking reveals what was there

Qubits: all four amplitudes at once

$|\psi\rangle = \sum_{ab} c_{ab}|ab\rangle = (c_{00}, c_{01}, c_{10}, c_{11})^\top$
measuring gives ab with probability $|c_{ab}|^2$

example: $|\psi\rangle = (\sqrt{0.1}, \sqrt{0.3}, \sqrt{0.4}, \sqrt{0.2})^\top$, probabilities $0.1 + 0.3 + 0.4 + 0.2 = 1$

Quantum Mechanics: Deterministic Evolution

Schrödinger equation: the law of motion of $|\psi\rangle$

$$\frac{\mathrm{d}}{\mathrm{d}t}|\psi\rangle = -i\,H\,|\psi\rangle \qquad \text{compare} \qquad \frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = \mathbf{F}(\mathbf{x})$$

the equation is linear in $|\psi\rangle$

The Hamiltonian H

the energy of the system written as a matrix; for two qubits a 4×4 matrix
diagonal: energy of each configuration;
off-diagonal: how strongly one configuration turns into another

$$H = \begin{pmatrix} E_{00} & g & g & 0 \\ g & E_{01} & 0 & g \\ g & 0 & E_{10} & g \\ 0 & g & g & E_{11} \end{pmatrix}$$

example: g flips one qubit

H plays the role of $\mathbf{F}$: given $|\psi(0)\rangle$, the evolution is **deterministic**

Quantum Mechanics: Truly Probabilistic Measurement

Measurement: Born rule and collapse of the wave function

$$P_a = |a\rangle\langle a|, \qquad p(a) = \langle\psi|P_a|\psi\rangle = |\langle a|\psi\rangle|^2, \qquad |\psi\rangle \xrightarrow{\text{outcome } a} \frac{P_a|\psi\rangle}{\sqrt{p(a)}}$$

Example: measure only the first of two qubits, $P_0 = |0\rangle\langle 0| \otimes I$

$$|\psi\rangle = c_{00}|00\rangle + c_{01}|01\rangle + c_{10}|10\rangle + c_{11}|11\rangle$$

$$\text{outcome 0 with } p(0) = |c_{00}|^2 + |c_{01}|^2\text{:} \quad |\psi\rangle \to \frac{c_{00}|00\rangle + c_{01}|01\rangle}{\sqrt{p(0)}}$$

$$\text{example } |\psi\rangle = (\sqrt{0.1}, \sqrt{0.3}, \sqrt{0.4}, \sqrt{0.2})^\top\text{: } p(0) = 0.4,\ |\psi\rangle \to \tfrac{1}{2}|00\rangle + \tfrac{\sqrt{3}}{2}|01\rangle$$

the outcome is random, and **the measurement changes the state**: c_{10}, c_{11} are gone; repeating it gives the same outcome

M. Born, Z. Phys. 37, 863 (1926); J. von Neumann (1932)

Forecasting in Practice

Classical deterministic systems

This lecture treats classical systems with deterministic laws $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$, observed with incomplete information.

Even in a deterministic classical world we cannot forecast from the laws alone:

- the full state $\mathbf{x}$ is never measured, only a few numbers
- the law $\mathbf{F}$ is known exactly only for simple systems
- chaos amplifies every small error in $\mathbf{x}(t_0)$ and in $\mathbf{F}$

What we have: reduced information about the system

a few numbers measured at discrete time steps, $\{s_1, s_2, \ldots, s_k\}$ (made precise as a **time series** below). The question of this lecture: how to forecast $s_{k+\tau}$ from it?

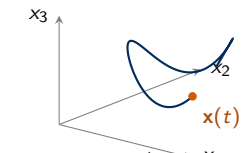
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Dynamical Systems and Time Series



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Phase Space, Flows and Maps



state $\mathbf{x}(t) \in \mathbb{R}^d$, a point moving in phase space

Phase space

The set of all possible states $\mathbf{x} \in \mathbb{R}^d$: each point is one complete state, the motion is a trajectory through it.

Flow and map

$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$: flow, continuous time

$\mathbf{x}_{n+1} = \mathbf{f}(\mathbf{x}_n)$: map, discrete steps

- flow: e.g. a pendulum, Lorenz-63; map: e.g. the logistic map $x_{n+1} = r\,x_n(1 - x_n)$
- a flow sampled every Δt is a map, $\mathbf{x}_{k+1} = \varphi_{\Delta t}(\mathbf{x}_k)$: what a stored record sees
- **chaos**: bounded, never repeating, nearby states separate exponentially

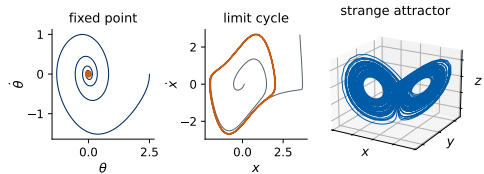
S. H. Strogatz, *Nonlinear Dynamics and Chaos*, 2nd ed., Westview Press (2015)

Flow Map and Attractors

Flow map of $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$

$$\varphi_t : \mathbf{x}_0 \mapsto \mathbf{x}(t; \mathbf{x}_0), \qquad \varphi_0 = \mathrm{id}, \quad \varphi_{t+s} = \varphi_t \circ \varphi_s, \quad \varphi_{-t} = \varphi_t^{-1}$$

sampled every Δt: $\mathbf{x}_{k+1} = \varphi_{\Delta t}(\mathbf{x}_k)$

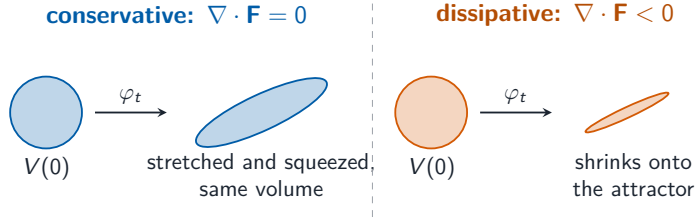


Attractor

the set A where the motion ends up after the transient, for all starts in its **basin**

fixed point: pendulum at rest; limit cycle: Van der Pol; strange (Lorenz-63): never closes, fractal dimension 2.06

Phase-Space Volume: Conserved or Shrinking



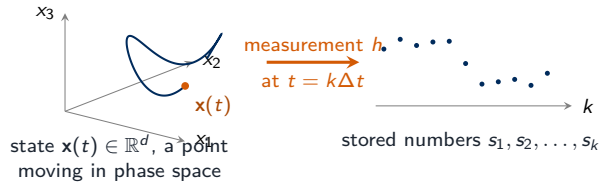
Volume of a cloud of initial states

$$\frac{\mathrm{d}V}{\mathrm{d}t} = \int_{A_t} \nabla \cdot \mathbf{F} \, \mathrm{d}^d x$$

$\nabla \cdot \mathbf{F} = \sum_i \partial F_i / \partial x_i$: divergence, how fast the flow spreads out

- Hamiltonian (frictionless pendulum, planets): energy and volume conserved
- dissipation (friction, viscosity): energy lost, volume shrinks; Lorenz-63: $\nabla \cdot \mathbf{F} = -41/3$

Measured Time Series



Time series

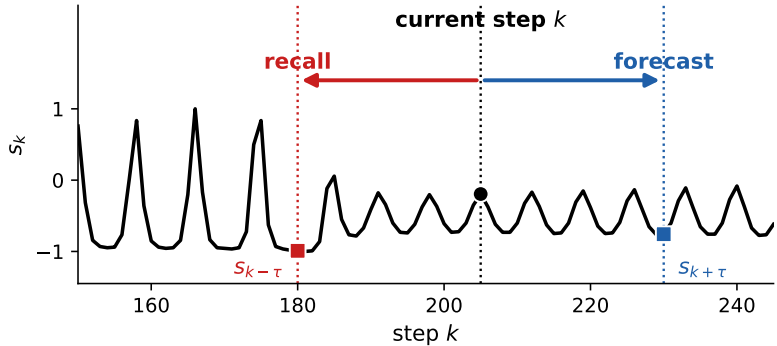
$$s_k = h(\mathbf{x}(k\Delta t)) + \eta_k, \quad k = 1, \ldots, N$$

h: measurement function; η_k: measurement noise, corrupts the record only; Δt: sampling interval

A real system is seen only through a projection

The trajectory $\mathbf{x}(t) \in \mathbb{R}^d$ in phase space is hidden. A measurement records one number per step, $h : \mathbb{R}^d \to \mathbb{R}$, a projection of the full state.

Recall and Forecast Tasks



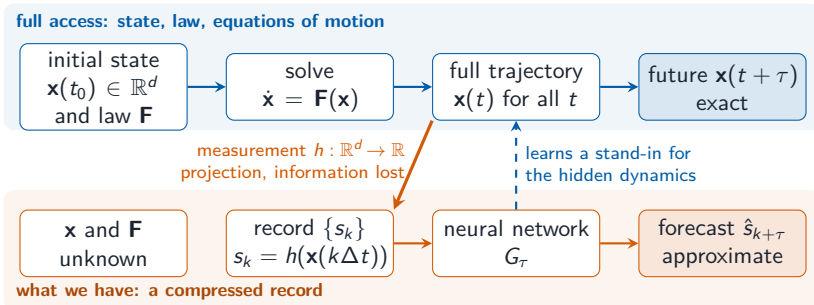
Task: from the stored $s_1, \ldots, s_k$ alone infer **forecast** $s_{k+\tau}$ (τ steps ahead, the horizon) or **recall** $s_{k-\tau}$ (τ steps back, tests memory).

without access to the state $\mathbf{x}$ or the law $\mathbf{F}$.

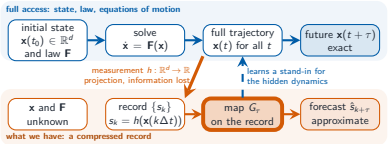
Full Access and the Compressed Record

The law $\mathbf{F}$ in $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$

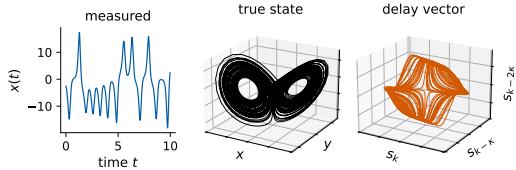
contains all effects: forces, couplings, friction, driving, delays, parameters. Known well for a pendulum, a circuit, planetary orbits; only approximately for cells, ecosystems, climate (missing terms, uncertain parameters).



Takens: a Picture of the Hidden Evolution



G_τ acts on the compressed record only



Lorenz-63: measured $x(t)$, hidden attractor, picture rebuilt from x

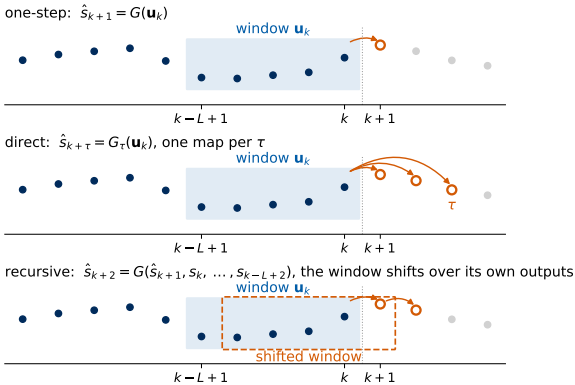
When delay coordinates determine a state

delay vector $\mathbf{u}_k = (s_k,\ s_{k-\kappa},\ \ldots,\ s_{k-(m-1)\kappa}) \in \mathbb{R}^m$, built from the record only.
Delay vectors of one measured variable reconstruct the state when m is large enough (Takens 1981).

The forecast $\hat{s}_{k+\tau} = G_\tau(\mathbf{u}_k)$ then has a single-valued target.

F. Takens, Lect. Notes Math. 898, 366 (1981); T. Sauer *et al.*, J. Stat. Phys. 65, 579 (1991)

Forecast, Horizon and Window



J. D. Farmer, J. J. Sidorowich, Phys. Rev. Lett. 59, 845 (1987)

Forecast at horizon τ

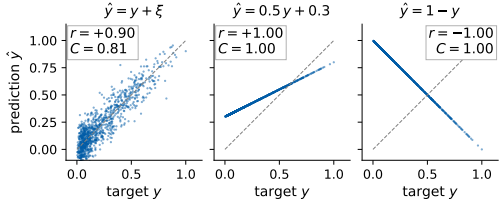
$$\hat{s}_{k+\tau} = G_\tau(\mathbf{u}_k)$$
$$\mathbf{u}_k = (s_k, \ldots, s_{k-L+1})$$

G_τ: our model for the value τ steps ahead (recall: back); $\mathbf{u}_k$: last L values, a stand-in for $\mathbf{x}$

Aim

1. a window prescription: spacing κ and number m of past values sufficient to predict the observable
2. a model G_τ learned from the record

Measure of Forecast Quality



$\hat{y} = \hat{s}_{k+\tau}$ against $y = s_{k+\tau}$; dashed: $\hat{y} = y$

- ideal: all points on the diagonal $\hat{y} = y$
- C: fit to a straight line; noise lowers it, scale and offset do not
- $\hat{y} = 1 - y$ also has $C = 1$: check the sign of r

Pearson correlation and capacity

$$r(\tau) = \frac{\sum_i (\hat{y}_i - \bar{\hat{y}})(y_i - \bar{y})}{\sqrt{\sum_i (\hat{y}_i - \bar{\hat{y}})^2 \sum_i (y_i - \bar{y})^2}}, \qquad C(\tau) = r(\tau)^2, \qquad C_{\text{tot}} = \sum_{\tau=1}^{50} C(\tau)$$

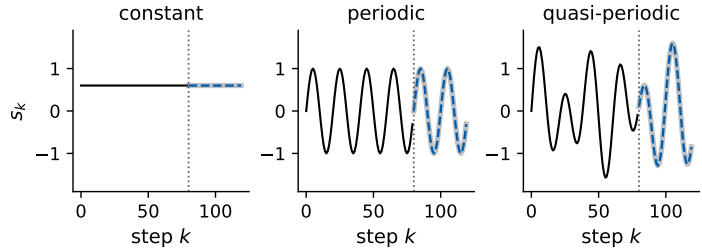
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Regular and Chaotic Signals



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Regular Signals



Past (black) up to $k = 80$, true continuation (grey) and forecast (dashed blue).

- **constant**: the last value, $\hat{s}_{k+\tau} = s_k$ (persistence)
- **periodic** with period P: the value one period earlier, $\hat{s}_{k+\tau} = s_{k+\tau-P}$
- **quasi-periodic**, two sines with an irrational frequency ratio: never repeats, yet obeys an exact linear recurrence $s_k = \sum_{j=1}^{4} a_j s_{k-j}$, fitted on the past and iterated

All three forecasts are exact to round-off, these are trivial tasks.

From Navier–Stokes to Lorenz-63

Fluid heated from below: Navier–Stokes and heat equations

$$\partial_t \mathbf{v} + (\mathbf{v}\cdot\nabla)\mathbf{v} = -\tfrac{1}{\rho_0}\nabla p + \nu\nabla^2\mathbf{v} + g\alpha\, T\,\hat{\mathbf{z}}, \qquad \nabla\cdot\mathbf{v} = 0, \qquad \partial_t T + (\mathbf{v}\cdot\nabla)T = \chi\nabla^2 T$$

$\mathbf{v}$: velocity; T: temperature relative to a fixed reference; ν: viscosity; χ: thermal diffusivity; $g\alpha T$: buoyancy

- the state is a field, a value at every point $\mathbf{r}$: infinite-dimensional phase space
- $(\mathbf{v}\cdot\nabla)\mathbf{v}$ makes the law nonlinear: weather, oceans, turbulence
- Saltzman (1962): Fourier modes; Lorenz (1963) kept only three, x, y, z
- even this caricature of the true $\mathbf{F}$ is chaotic: the limit of weather forecasts

B. Saltzman, J. Atmos. Sci. 19, 329 (1962); E. N. Lorenz, J. Atmos. Sci. 20, 130 (1963)

From Navier–Stokes to Lorenz-63 in Five Steps

1. **2D rolls**: flow in the (x, z) plane from a stream function $\Upsilon(x, z, t)$, so $\nabla \cdot \mathbf{v} = 0$
2. **temperature**: T = linear conduction profile + $\theta(x, z, t)$, the change caused by convection
3. **Fourier modes**: Υ, θ as sums of spatial patterns with amplitudes depending on t: infinitely many ODEs (Saltzman)
4. **keep three modes** (Lorenz), with roll width H/a

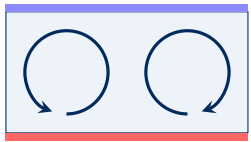
$$\Upsilon \propto X(t) \sin \tfrac{\pi a x}{H} \sin \tfrac{\pi z}{H}, \qquad \theta \propto Y(t) \cos \tfrac{\pi a x}{H} \sin \tfrac{\pi z}{H} - Z(t) \sin \tfrac{2\pi z}{H}$$

5. **project and drop all other modes**:
   $\dot{X} = \sigma(Y - X), \quad \dot{Y} = X(\rho - Z) - Y, \quad \dot{Z} = XY - \beta Z$

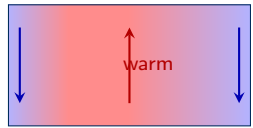
   $\sigma = \nu/\chi$; $\rho = \mathrm{Ra}/\mathrm{Ra}_c$, heating over its onset value; $\beta = 4/(1 + a^2) = 8/3$ for $a^2 = 1/2$

B. Saltzman, J. Atmos. Sci. 19, 329 (1962); E. N. Lorenz, J. Atmos. Sci. 20, 130 (1963)

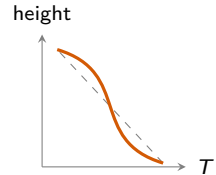
The Three Modes: What X, Y, Z Mean



X: speed of the rolls
sign = sense of rotation



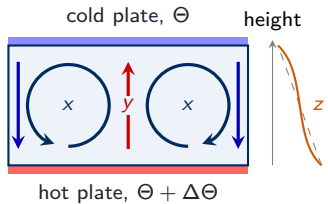
Y: temperature difference
rising vs sinking fluid



Z: bending of the
vertical profile (mixing)

- a Fourier mode is a fixed spatial pattern of the fluid; its amplitude $X(t)$, $Y(t)$, $Z(t)$ says how strongly it is present
- the true field needs infinitely many modes: smaller rolls, finer temperature structure
- Lorenz kept the three largest patterns: $\mathbf{x} = (X, Y, Z)$ is a three-number caricature of the whole fluid

Lorenz-63 Model



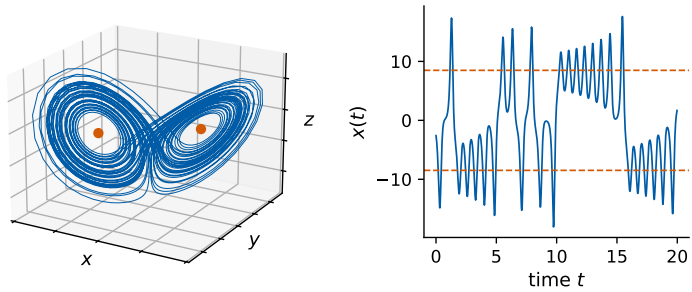
Three Fourier modes of the convecting fluid, $\mathbf{x} = (x, y, z)$:

$$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}), \quad \mathbf{F}(\mathbf{x}) = \begin{pmatrix} \sigma(y - x) \\ x(\rho - z) - y \\ xy - \beta z \end{pmatrix}$$

- x: speed of the convection rolls, its sign the sense of rotation
- y: temperature difference between rising (red) and sinking (blue) fluid
- z: bending of the vertical temperature profile (orange) away from pure conduction (dashed)
- $\sigma = 10$: Prandtl number, viscosity over thermal diffusivity of the fluid
- $\rho = 28$: heating, the temperature difference of the plates in units of its value at the onset of convection
- $\beta = 8/3$: geometric factor, set by the width-to-height ratio of the rolls

E. N. Lorenz, J. Atmos. Sci. 20, 130 (1963)

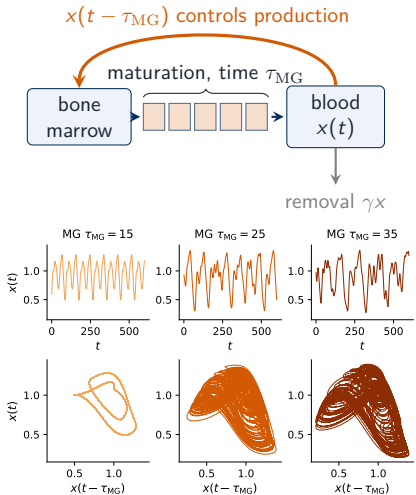
Lorenz-63 Strange Attractor



$\rho = 28$; left: trajectory in (x, y, z), dots: fixed points of the two steady rolls; right: $x(t)$, dashed: x at the fixed points

- $\rho < 1$ rest; $1 < \rho < 24.74$ steady rolls; $\rho = 28$ the rolls flip irregularly
- stored signal $s_k = x(k\Delta t)$, $\Delta t = 0.02$

Mackey–Glass Model



Production follows the count of τ_{MG} ago:

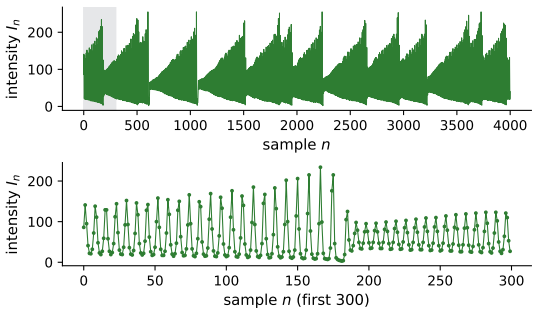
$$\dot{x} = \underbrace{\beta \frac{x(t-\tau_{\mathrm{MG}})}{1+x(t-\tau_{\mathrm{MG}})^{10}}}_{\text{production}} - \underbrace{\gamma x}_{\text{removal}}$$

x: blood-cell concentration; $\beta = 0.2$, $\gamma = 0.1$; panels: $x(t)$ (top), $x(t-\tau_{\mathrm{MG}})$ against $x(t)$ (bottom)

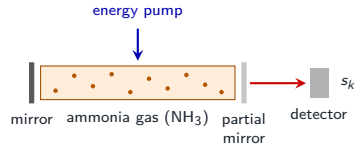
- $\tau_{\mathrm{MG}} = 15$: limit cycle, a closed line
- $\tau_{\mathrm{MG}} = 25, 35$: chaotic, the orbit never closes and fills a band

M. C. Mackey, L. Glass, Science 197, 287 (1977), doi:10.1126/science.267326

Santa Fe Laser



Intensity of a far-infrared ammonia laser.

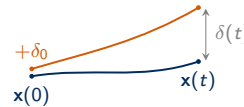


- field, polarisation and inversion obey the Lorenz equations, the pump playing ρ
- stored energy leaves in flashes that grow and collapse
- measured 8-bit record (Santa Fe competition)

H. Haken, Phys. Lett. A 53, 77 (1975); U. Hübner *et al.*, Phys. Rev. A 40, 6354 (1989)

Propagation of Small Errors

Equations of motion $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$; second trajectory $\tilde{\mathbf{x}}(t) = \mathbf{x}(t) + \delta(t)$, $\delta(0) = \delta_0$.



Linearisation

$$\dot{\delta} = \mathbf{F}(\mathbf{x}+\delta) - \mathbf{F}(\mathbf{x}) \approx J(\mathbf{x}(t))\,\delta, \quad J_{ij} = \frac{\partial F_i}{\partial x_j} \qquad \text{first order: linear in } \delta$$

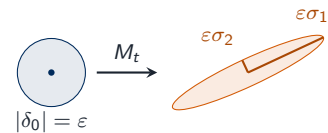
$$\delta(t) = M_t\,\delta_0, \quad (M_t)_{ij} = \frac{\partial x_i(t)}{\partial x_j(0)} \qquad \text{change of } x_i(t) \text{ per change of } x_j(0)$$

$$\frac{\mathrm{d}}{\mathrm{d}t}(M_t)_{ij} = \sum_k J_{ik}(\mathbf{x}(t))\,(M_t)_{kj}, \quad (M_0)_{ij} = \delta_{ij} \qquad \text{same linear equation, } M_0 = I$$

- M_t: Jacobian of the flow map, as for the volume; carries a small error from 0 to t

Growth Factors of the Error

The growth of a small error depends on its direction. The eigenproblem below gives the directions and their growth factors.



Growth factors

$$|\delta(t)|^2 = \delta(t)^\top \delta(t) = \delta_0^\top M_t^\top M_t\, \delta_0 \qquad \text{squared length of the error}$$

$$\tilde{M}_t \coloneqq M_t^\top M_t \qquad \text{symmetric } d \times d \text{ matrix}$$

$$\tilde{M}_t\, \mathbf{v}_i = \sigma_i^2\, \mathbf{v}_i, \quad \sigma_1 \geq \cdots \geq \sigma_d \geq 0 \qquad \text{eigenproblem, } \mathbf{v}_i \text{ orthonormal}$$

$$\delta_0 = \varepsilon \mathbf{v}_i \;\Rightarrow\; |\delta(t)| = \varepsilon\, \sigma_i \qquad \sigma_i\text{: growth factor along } \mathbf{v}_i$$

- a ball of errors becomes an ellipsoid with semi-axes $\varepsilon\sigma_1, \ldots, \varepsilon\sigma_d$
- the state stays bounded, only the error grows

Lyapunov Exponents

Lyapunov exponents: growth rates of the error axes

$$\lambda_i = \lim_{t\to\infty} \frac{1}{t} \ln \sigma_i(t), \qquad \sigma_i(t) \sim e^{\lambda_i t}, \qquad i = 1, \dots, d$$

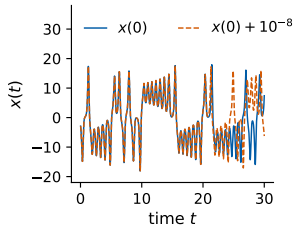
- bounded nonperiodic motion with $\lambda_1 > 0$: **chaos**; the exponents are the same for almost every starting point on the attractor
- for a flow, one exponent is zero (the direction along the orbit)
- sum rule: $\lambda_1 + \cdots + \lambda_d = \langle \nabla \cdot \mathbf{F} \rangle$, the average volume growth rate, < 0 for a dissipative system

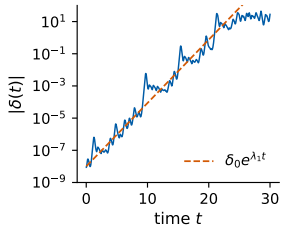
Lorenz-63, measured

$\lambda_1 = 0.907, \quad \lambda_2 = 0.0001, \quad \lambda_3 = -14.573, \qquad \sum_i \lambda_i = -13.667 \quad (\nabla \cdot \mathbf{F} = -41/3)$

V. I. Oseledets, Trans. Moscow Math. Soc. 19, 197 (1968); G. Benettin *et al.*, Meccanica 15, 9 (1980)

Sensitivity to Initial Conditions





E. N. Lorenz,
J. Atmos. Sci. 20, 130 (1963)

Lorenz-63 from two states 10^{-8} apart; left $x(t)$ of both, right their distance $|\delta(t)|$.

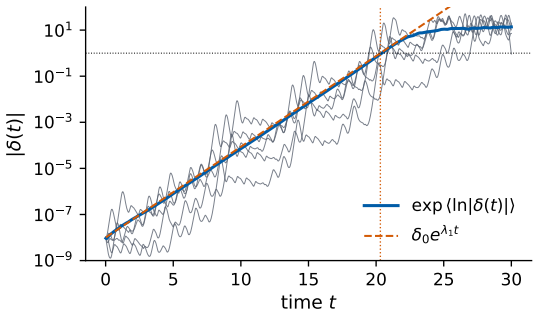
Largest Lyapunov exponent

$$|\delta(t)| \approx |\delta_0|\, e^{\lambda_1 t}, \qquad \lambda_1 = \lim_{t\to\infty} \lim_{|\delta_0|\to 0} \frac{1}{t} \ln \frac{|\delta(t)|}{|\delta_0|}$$

$\delta(t)$: difference of the two states; $T_\lambda = 1/\lambda_1$: Lyapunov time

The pair agrees until $t \approx 20$; slope of $\ln|\delta|$ 0.93, average $\lambda_1 = 0.907$.

Fundamental Limit of the Prediction Horizon



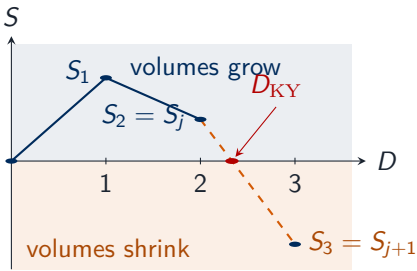
$|\delta(t)|$ of five pairs (grey) and $\exp\langle \ln|\delta|\rangle$ over 200 pairs (blue).

An initial error δ_0 grows until it reaches a tolerance Δ,

$$T_{\text{pred}} = \frac{1}{\lambda_1} \ln \frac{\Delta}{\delta_0}$$

- Lorenz, $|\delta_0| = 10^{-8}$: $\Delta = 1$ at $t = 20.4$, formula 20.3
- start 1000 times more precise, $\delta_0 \to \delta_0/1000$: only $\ln(1000)\, T_\lambda \approx 6.9\, T_\lambda$ longer
- chaos ($\lambda_1 > 0$): finite horizon even for the exact equations; a learned model stays below T_{pred}

Kaplan–Yorke Dimension



Schematic partial sums for a map. S_k is the maximal asymptotic growth rate of k-volumes.

J. L. Kaplan, J. A. Yorke, Lect. Notes Math. 730, 204 (1979)

Dimension from the spectrum

$$D_{\mathrm{KY}} = j + \frac{S_j}{|\lambda_{j+1}|}$$

j: last k with $S_k \geq 0$; straight line from S_j to S_{j+1} crosses zero at D_{KY}

- stable fixed point 0 and stable limit cycle 1. Chaos can have integer dimension.
- Lorenz-63: $j = 2$, $D_{\mathrm{KY}} = 2.06$
- a dimension estimate from the spectrum, with no general equality to box dimension

3

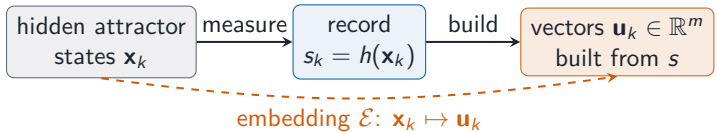
Delay Embedding: a Faithful Picture of the Attractor

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Goal: a Faithful Picture of the Attractor

What we want

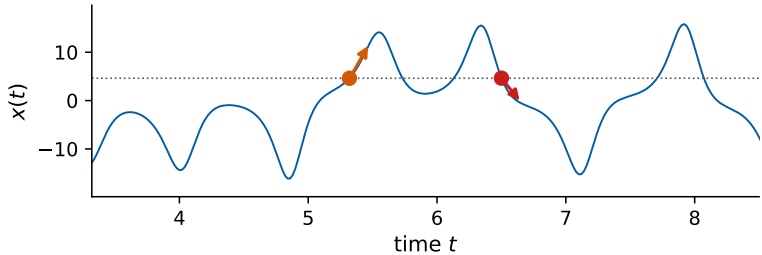
from the scalar record $s_1, s_2, \ldots$ build vectors $\mathbf{u}_k$ whose cloud is a **faithful copy of the attractor**: an **embedding** of the dynamics



Faithful means

1. one-to-one: different states, different $\mathbf{u}_k$
2. smooth: neighbours stay neighbours
3. dynamics carry over: $\mathbf{u}_k$ fixes the future

One Value Does Not Fix the State



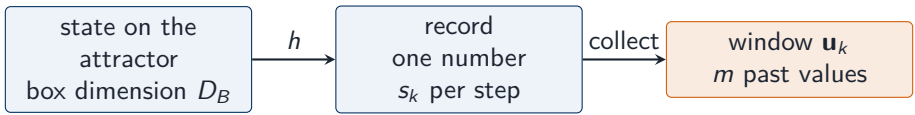
Lorenz-63 $x(t)$; two moments with the same value (dotted), one rising, one falling

- state: a point $\mathbf{x} = (x, y, z)$ in phase space; the equations of motion fix its future
- we measure $s_k = x$, the projection of the point on one axis
- same x, different y, z: different points, different futures; a forecast of s_{k+1} needs past values $s_k, s_{k-\kappa}, \ldots$

From Dimension to Measurements

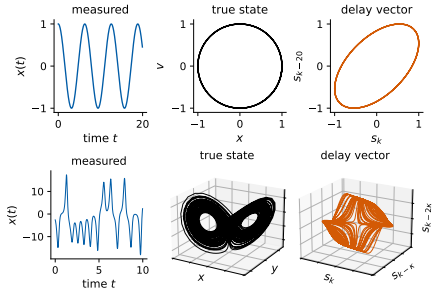
Aim

number m and spacing κ of past values such that the window $\mathbf{u}_k = (s_k, s_{k-\kappa}, \ldots, s_{k-(m-1)\kappa})$ fixes the state $\mathbf{x}_k$, and with it every future value $s_{k+\tau}$



- Lorenz has three state coordinates; only x is measured
- one measurement per step is one number; past values add the missing ones, because the dynamics wrote y, z into the later x
- $m > 2D_B$ is sufficient under the fractal embedding hypotheses; D_{KY} is only a guide

Phase Space from One Coordinate



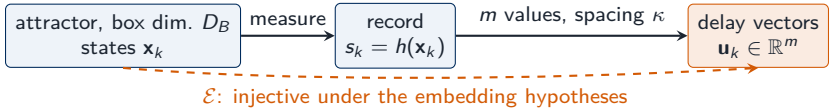
True state and delay vector

$\mathbf{x}_k = \big(x, y, z\big)(k\Delta t)$ three different quantities at one time

$\mathbf{u}_k = \big(s_k, s_{k-\kappa}, s_{k-2\kappa}\big), \quad s_k = x(k\Delta t)$ one quantity at three times

$\mathbf{u}_k \neq \mathbf{x}_k$, and can carry state information: Lorenz-63 with $\kappa = 10$ rebuilds both lobes; oscillator: $v_k \approx (x_k - x_{k-1})/\Delta t$ is hidden in the past

Delay Map



Delay map $\mathcal{E}$: state $\to$ vector in $\mathbb{R}^m$

$$\mathcal{E}(\mathbf{x}) := \big(h(\mathbf{x}),\ h(\varphi_{-\kappa\Delta t}(\mathbf{x})),\ \ldots,\ h(\varphi_{-(m-1)\kappa\Delta t}(\mathbf{x}))\big) \qquad \text{measure at } 0, -\kappa, -2\kappa, \ldots \text{ steps}$$

$$\mathbf{x}_{k-\kappa} = \varphi_{-\kappa\Delta t}(\mathbf{x}_k) \ \Rightarrow\ h(\varphi_{-\kappa\Delta t}(\mathbf{x}_k)) = h(\mathbf{x}_{k-\kappa}) = s_{k-\kappa} \qquad \text{flow map backwards}$$

$$\mathcal{E}(\mathbf{x}_k) = \big(s_k, s_{k-\kappa}, \ldots, s_{k-(m-1)\kappa}\big) = \mathbf{u}_k \qquad \text{from the record}$$

h: measurement function, state $\to$ one stored number; κ: delay; m: dimension

- $\mathbf{u}_k$: numbers computed from the record; equal to $\mathcal{E}$ of the hidden true state $\mathbf{x}_k$
- $\mathcal{E}$, h, φ are never needed explicitly: the theorem is about $\mathcal{E}$, the data give $\mathbf{u}_k$

Takens Embedding Theorem

Two sufficient conditions

Smooth compact manifold: $m \geq 2d + 1$ gives a smooth embedding for generic dynamics and measurements; d is the manifold dimension.
Compact fractal set: $m > 2D_B$ gives generic injectivity under periodic-orbit conditions; D_B is the box dimension.

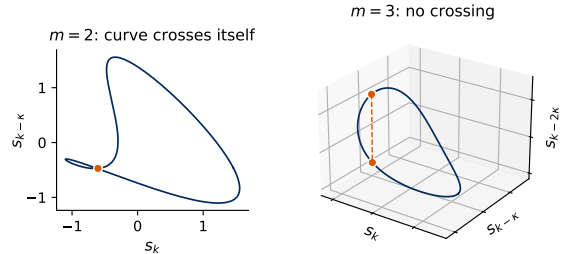
- the delay map is one-to-one, so each delay vector fixes one state
- for a smooth manifold the map is smooth in both directions, so the lobes and holes of the attractor are kept
- in the smooth case the Lyapunov exponents are preserved

The reconstructed dynamics follows from the true flow $\varphi_{\Delta t}$,

$$\mathbf{u}_{k+1} = \left(\mathcal{E} \circ \varphi_{\Delta t} \circ \mathcal{E}^{-1}\right)(\mathbf{u}_k).$$

F. Takens, Lect. Notes Math. 898, 366 (1981); T. Sauer *et al.*, J. Stat. Phys. 65, 579 (1991)

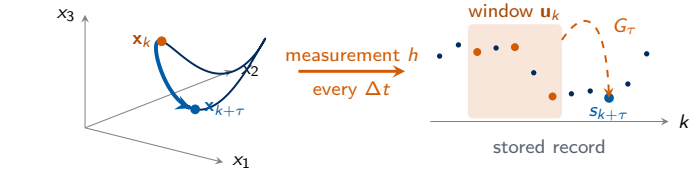
A Sufficient Embedding Dimension: $m > 2D_B$



$s(t) = \cos t + 0.6\cos(2t + 0.7)$: closed curve, $D_B = 1$; orange: two different states

- $m = 2$: two states, one delay vector: no $\mathcal{E}^{-1}$, future ambiguous
- $m = 3 > 2D_B$: one state per delay vector: $\mathcal{E}^{-1}$ exists, one future
- two pieces of dimension D in $\mathbb{R}^m$ meet in general only if $D + D \geq m$
- often fewer suffice; D_{KY} and FNN are guides for choosing m

Forecast Map from Delay Vectors



Forecast map

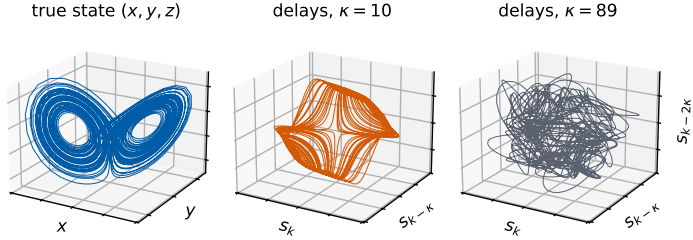
$$G_\tau = h \circ \varphi_{\tau\Delta t} \circ \mathcal{E}^{-1}, \qquad \hat{s}_{k+\tau} = G_\tau(\mathbf{u}_k)$$

$\varphi_{\tau\Delta t}$: flow map of $\mathbf{F}$, solve $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$ for time $\tau\Delta t$: $\varphi_{\tau\Delta t}(\mathbf{x}_k) = \mathbf{x}_{k+\tau}$

- $\mathcal{E}$ unknown (contains h and the flow); we only compute $\mathbf{u}_k$ from the record
- $\mathcal{E}^{-1}$ exists, so G_τ exists; learned without knowing h, φ, $\mathcal{E}$; window $L = (m-1)\kappa + 1$

Choosing the Delay κ

Aim: the spacing κ of the window $\mathbf{u}_k = (s_k, s_{k-\kappa}, \ldots, s_{k-(m-1)\kappa})$, from the record alone.

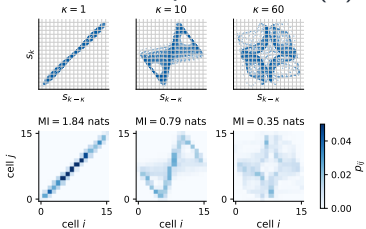


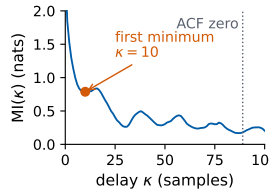
Lorenz-63: true state (x, y, z) and delay vectors $(s_k, s_{k-\kappa}, s_{k-2\kappa})$, $s_k = x(k\Delta t)$

- κ too small: $s_{k-\kappa} \approx s_k$, all points near the diagonal
- $\kappa = 10$: the two lobes reappear, smoothly deformed
- $\kappa = 89$ (first zero of the autocorrelation): the lobes are smeared out; zero correlation does not mean independence

Mutual Information

Aim: a number that picks κ: $\mathrm{MI}(\kappa)$, how much $s_{k-\kappa}$ tells about s_k.





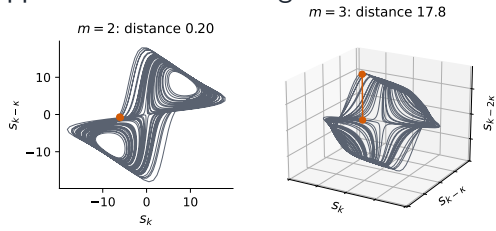
Histogram estimate of mutual information

16×16 histogram; same $N - 100$ pairs per lag ($\kappa_{\max} = 100$): $p_{ij} = \dfrac{n_{ij}}{N - \kappa_{\max}}$, $p_i = \sum_j p_{ij}$, $p_j = \sum_i p_{ij}$, $\mathrm{MI}(\kappa) = \sum_{i,j} p_{ij} \ln \dfrac{p_{ij}}{p_i p_j}$. Population MI vanishes for independence. Sampling can give positive estimates even then. The first minimum is a heuristic.

A. M. Fraser, H. L. Swinney, Phys. Rev. A 33, 1134 (1986)

False Nearest Neighbours (FNN)

Aim: a useful number m of delay coordinates. The bound $m > 2D_B$ needs an unknown box dimension; FNN supplies a finite-data diagnostic.



Lorenz x, delay vectors with $m = 2$ and $m = 3$ coordinates; orange: one pair of points

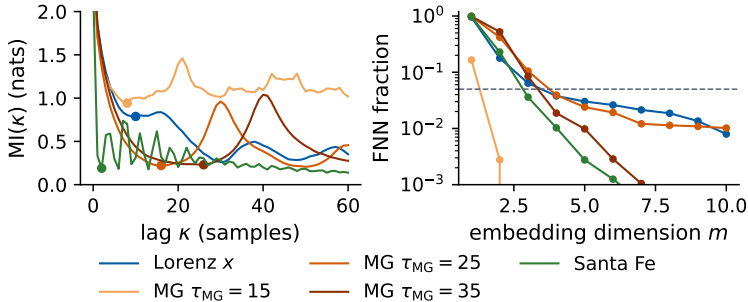
- close in $m = 2$, far in $m = 3$: a **false neighbour**, the attractor crossing itself
- different states, different futures: neighbours give the wrong forecast

M. B. Kennel, R. Brown, H. D. I. Abarbanel, Phys. Rev. A 45, 3403 (1992)

Determining m with the FNN Criterion

1. κ fixed from MI; for a trial m build every delay vector $\mathbf{u}_k^{(m)} = (s_k, s_{k-\kappa}, \ldots, s_{k-(m-1)\kappa})$
2. nearest neighbour $\mathbf{u}_j^{(m)}$: smallest distance $r_k = \|\mathbf{u}_k^{(m)} - \mathbf{u}_j^{(m)}\| = \big(\sum_{i=0}^{m-1}(s_{k-i\kappa} - s_{j-i\kappa})^2\big)^{1/2}$, searched among $|k-j| > 50$, so that j is not the next sample of the same pass
3. add one coordinate, $m \to m+1$: the pair gets $s_{k-m\kappa}$ and $s_{j-m\kappa}$, a new difference $\Delta_k = |s_{k-m\kappa} - s_{j-m\kappa}|$
4. **false neighbour** if $\Delta_k > 15\, r_k$: pulled apart far beyond their distance (Kennel *et al.* add an absolute test, $\sqrt{r_k^2 + \Delta_k^2} > 2\sigma_s$, compared in the two-test table below)
5. $\mathrm{FNN}(m)$: fraction of k with a false neighbour; increase m until $\mathrm{FNN}(m) < 5\,\%$, an empirical estimate of m

Dimension from the FNN Fraction



left: $\mathrm{MI}(\kappa)$, dots at the chosen κ; right: fraction of false neighbours against m, dashed: 5 %

- empirical m: first below 5 %: Lorenz 4, MG 15 2, Santa Fe 3
- suggested window span $L = (m-1)\kappa + 1$

Ratio Test and Absolute Test

Ratio: $\Delta_k > 15 r_k$ **Absolute:** $\sqrt{r_k^2 + \Delta_k^2} > 2\sigma_s$

r_k: neighbour distance; Δ_k: added-coordinate separation; σ_s: record standard deviation

record	κ	ratio $m = 8$	combined $m = 4$	combined $m = 8$
Lorenz x	10	0.019	0.040	0.040
Mackey–Glass 25	16	0.011	0.049	0.055
Gaussian noise	2	0.000	0.174	0.275
central England anomaly	12	0.000	0.169	0.250

Theiler windows: Lorenz and MG 50; Gaussian control 20. Gaussian ratio first vanishes at $m = 7$.

- ratio alone becomes small for all four records as neighbour distances grow
- combined tests stay much lower for the deterministic benchmarks; noise and weather retain distant neighbours

M. B. Kennel *et al.*, Phys. Rev. A 45, 3403 (1992)

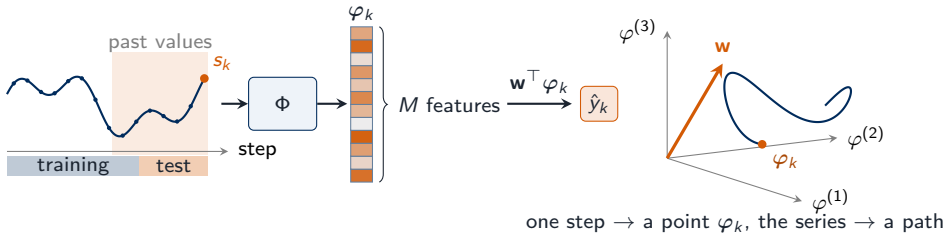
4

Linear Models: AR, MA and ARMA

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Features and Linear Readout

$$(s_1, \ldots, s_k) \xrightarrow{\Phi} \varphi_k \in \mathbb{R}^M \xrightarrow{\text{linear}} \hat{y}_k = \mathbf{w}^\top \varphi_k$$



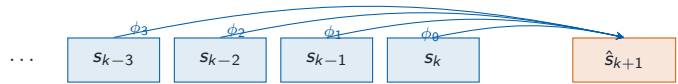
- Φ: M features of the last L values or a state $\mathbf{h}_k = U(\mathbf{h}_{k-1}, s_k)$; linear (the past values) or nonlinear (products, tanh units, a physical system)
- in AR and NVAR Φ is fixed and only the readout $\mathbf{w} \in \mathbb{R}^M$ is trained, by least squares
- recurrent neural networks train Φ and $\mathbf{w}$ together

Autoregressive Model AR(p)

AR(p)

$$\hat{s}_{k+1} = c + \sum_{i=0}^{p-1} \phi_i \, s_{k-i}$$

ϕ_i: weight of the value i steps back; c: offset; p: order



Fitting: one least-squares problem

$$(\hat{c}, \hat{\phi}_0, \ldots, \hat{\phi}_{p-1}) = \arg\min \sum_k \left(s_{k+1} - c - \sum_{i=0}^{p-1} \phi_i \, s_{k-i} \right)^2$$

linear in the parameters: one linear system, no gradient descent

G. E. P. Box, G. M. Jenkins, *Time Series Analysis: Forecasting and Control*, Holden-Day (1970)

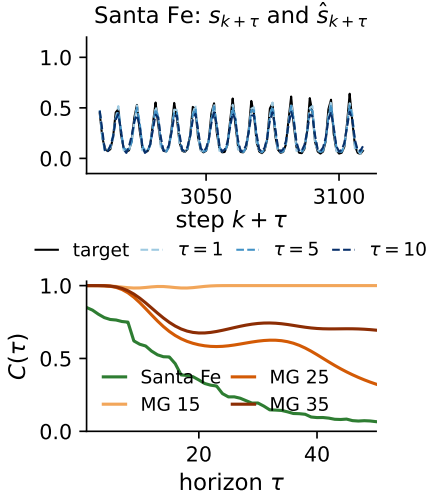
AR Forecast τ Steps Ahead

Direct AR

$$\hat{s}_{k+\tau} = c_\tau + \sum_{i=0}^{L-1} w_{\tau,i}\, s_{k-i}$$

c_τ: trainable offset, absorbs the mean of the signal; $w_{\tau,i}$: trainable weight of s_{k-i}; one set $(c_\tau, w_{\tau,0}, \ldots, w_{\tau,L-1})$ per horizon τ, by least squares on the training block; L on the validation block

- Mackey–Glass: $C(\tau)$ stays high over many steps
- Santa Fe: C falls within a few steps, the collapses are missed

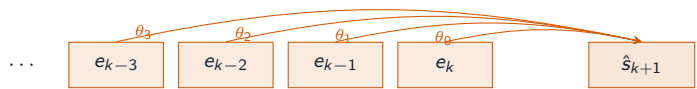


Moving-Average Model MA(q)

MA(q)

$$\hat{s}_{k+1} = \mu + \sum_{j=0}^{q-1} \theta_j \, e_{k-j}, \qquad e_k = s_k - \hat{s}_k$$

e_{k-j}: past prediction errors; θ_j: weight of the error j steps back; μ: mean of the signal on the training block; q: order



- the forecast is the mean plus a correction from the recent surprises e_k
- a shock e_k affects the next q forecasts only, after that the forecast returns to μ

G. E. P. Box, G. M. Jenkins, *Time Series Analysis: Forecasting and Control*, Holden-Day (1970)

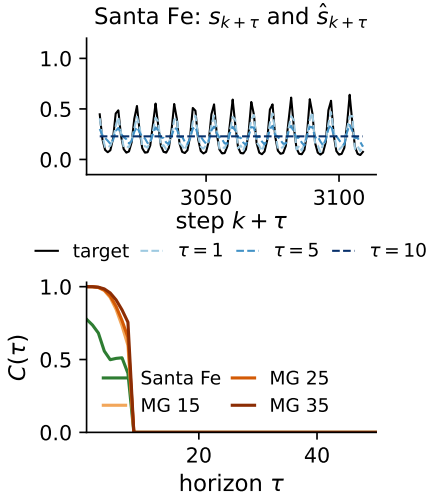
MA Forecast τ Steps Ahead

MA(q)

$$\hat{s}_{k+1} = \mu + \sum_{j=0}^{q-1} \theta_j \, e_{k-j}, \qquad e_k = s_k - \hat{s}_k$$

μ: mean of the signal over the whole training block, not of the window, fitted with θ_j; e_k: past prediction errors; θ_j: weights fitted on the training block; τ steps: iterate with future errors set to 0

- beyond q steps the forecast is the mean μ: no memory of the values
- weakest model on every signal

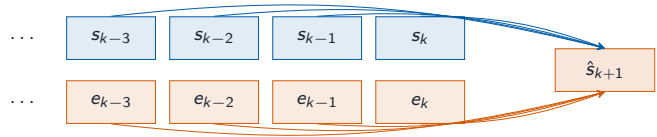


ARMA(p, q) Model

ARMA(p, q)

$$\hat{s}_{k+1} = c + \sum_{i=0}^{p-1} \phi_i \, s_{k-i} + \sum_{j=0}^{q-1} \theta_j \, e_{k-j}$$

both sums: p past values and q past errors



- the forecast is linear in the past values and the past errors e_k; the constant c absorbs the mean. Parameters and orders are fitted on the training data.

P. J. Brockwell, R. A. Davis, *Introduction to Time Series and Forecasting*, Springer (2016), doi:10.1007/978-3-319-29854-2

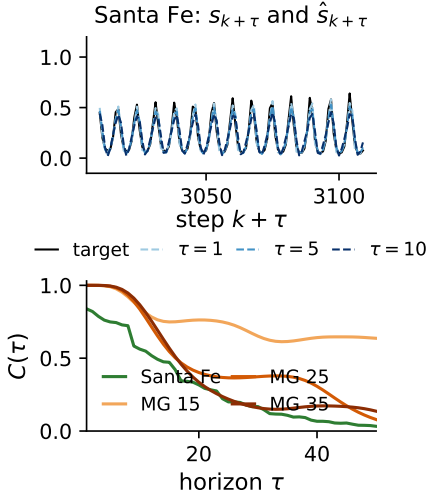
ARMA Forecast τ Steps Ahead

ARMA(p, q)

$$\hat{s}_{k+1} = c + \sum_{i=0}^{p-1} \phi_i \, s_{k-i} + \sum_{j=0}^{q-1} \theta_j \, e_{k-j}$$

c: constant; ϕ_i, θ_j: weights fitted on the training block; τ steps: the one-step model iterated, future errors set to 0

- close to AR: on these deterministic signals the MA part adds little
- iterated: errors of each step feed the next



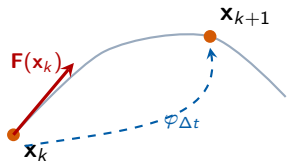
5

Nonlinear Vector Autoregression



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From the Velocity F to the Flow Map $\varphi_{\Delta t}$



red: velocity now, the tangent; blue: where the state is after Δt

$\mathbf{F}$: velocity now

$$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$$

direction and speed of the state at this instant

$\varphi_{\Delta t}$: position after Δt

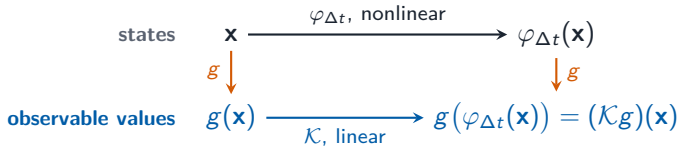
$$\varphi_{\Delta t}(\mathbf{x}) = \mathbf{x} + \int_0^{\Delta t} \mathbf{F}(\mathbf{x}(s))\,\mathrm{d}s \;\approx\; \mathbf{x} + \Delta t\,\mathbf{F}(\mathbf{x})$$

sum of velocities along the way; small Δt: Euler step

Properties of the flow map

$$\varphi_0 = \mathrm{id}, \qquad \varphi_{t+s} = \varphi_t \circ \varphi_s, \qquad \varphi_{-t} = \varphi_t^{-1}, \qquad \mathbf{x}_{k+n} = \varphi_{n\Delta t}(\mathbf{x}_k)$$

Koopman Operator: Nonlinear Dynamics as a Linear Map



Definition

observable $g : \mathbb{R}^d \to \mathbb{R}$

x_1, energy, $x_1 x_3$, $\sin x_2$; the measurement h

$$(\mathcal{K}_t g)(\mathbf{x}) := g(\varphi_t(\mathbf{x}))$$
$$\mathcal{K} := \mathcal{K}_{\Delta t}$$

B. O. Koopman, PNAS 17, 315 (1931)

Properties, inherited from φ_t

$$\mathcal{K}_0 = \mathrm{id}, \quad \mathcal{K}_t \mathcal{K}_s = \mathcal{K}_{t+s}, \quad g(\mathbf{x}_{k+n}) = (\mathcal{K}^n g)(\mathbf{x}_k)$$

linear for any $\mathbf{F}$: $\mathcal{K}(\alpha g_1 + \beta g_2)$

$$= \alpha\, g_1 \circ \varphi_{\Delta t} + \beta\, g_2 \circ \varphi_{\Delta t} = \alpha\, \mathcal{K} g_1 + \beta\, \mathcal{K} g_2$$

The Task: Find Observables g and a Matrix K

- $\mathcal{K}$ is linear, but acts on infinitely many functions: $\varphi_{\Delta t}$ may be nonlinear, the price is the dimension
- a finite set $\mathbf{g} = (g_1, \ldots, g_M)$ is useful if $\mathcal{K}$ maps it (nearly) into itself

The task

$$\mathbf{g}(\mathbf{x}_{k+1}) \approx K\,\mathbf{g}(\mathbf{x}_k) \quad \Rightarrow \quad \mathbf{g}(\mathbf{x}_{k+n}) \approx K^n\,\mathbf{g}(\mathbf{x}_k)$$

K: an $M \times M$ matrix fitted from data by least squares; forecasting becomes repeated multiplication by K

- a good choice of $\mathbf{g}$ makes this nearly exact; for some maps it is exact
- AR, NVAR and neural networks are three ways of choosing $\mathbf{g}$ from the record

Example: One Extra Observable Makes the Map Linear

Nonlinear map in (x, y)

$$x_{k+1} = a\,x_k, \qquad y_{k+1} = b\,y_k + c\,x_k^2$$

the term x_k^2 makes it nonlinear

$\mathcal{K}$ on three observables

$g_1 = x, \quad g_2 = y, \quad g_3 = x^2$

$$\mathcal{K} g_1 = x_{k+1} = a\,g_1$$
$$\mathcal{K} g_2 = y_{k+1} = b\,g_2 + c\,g_3$$
$$\mathcal{K} g_3 = x_{k+1}^2 = a^2\,g_3$$

each result is again a combination of g_1, g_2, g_3: the set closes

$\mathcal{K}$ as a 3×3 matrix

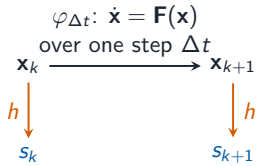
$$\begin{pmatrix} x \\ y \\ x^2 \end{pmatrix}_{k+1} = \underbrace{\begin{pmatrix} a & 0 & 0 \\ 0 & b & c \\ 0 & 0 & a^2 \end{pmatrix}}_{K} \begin{pmatrix} x \\ y \\ x^2 \end{pmatrix}_{k}$$

$\mathbf{g}_{k+n} = K^n\,\mathbf{g}_k$ exactly

rows of K: the coefficients on the left

for chaotic systems no finite set closes exactly: K is then only an approximation

Koopman and the Measured Series



hidden state above, measured values below

The measurement is an observable

$s_k = h(\mathbf{x}_k)$ is one instance of an observable, $g = h$:

$$(\mathcal{K}h)(\mathbf{x}_k) = h(\mathbf{x}_{k+1}) = s_{k+1}$$
$$s_{k+n} = (\mathcal{K}^n h)(\mathbf{x}_k)$$

- the time series is $\mathcal{K}$ acting again and again on one observable h
- forecasting = finding observables, built from the record, on which the evolution is (nearly) linear
- AR: past values $s_k, s_{k-1}, \ldots$; NVAR: past values and their products; neural networks: observables learned from data

Nonlinear Features of the Window

A linear readout is a flat plane through the data. To follow a curved dependence on the past we keep the window of past values and add their products as further features.

Nonlinear vector autoregression (NVAR)

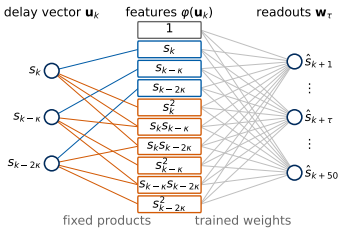
$$\varphi_k = \big(1,\ \underbrace{s_k, s_{k-1}, \ldots, s_{k-L+1}}_{\text{linear}},\ \underbrace{s_{k-i}\, s_{k-j}\ (0 \le i \le j \le L-1)}_{\text{quadratic}}\big), \qquad \hat{s}_{k+\tau} = \mathbf{w}_\tau^\top \varphi_k$$

- **trainable parameters**: the readout weights $\mathbf{w}_\tau$, one vector for every lead τ
- the features are fixed, so training is again one least-squares problem
- L and a stride κ (every κ-th value) are chosen by us

D. J. Gauthier, E. Bollt, A. Griffith, W. A. S. Barbosa, Nat. Commun. 12, 5564 (2021), doi:10.1038/s41467-021-25801-2

NVAR: Features and Loss

For $L = 2$: $\hat{s}_{k+1} = w_0 + \underbrace{w_1 s_k + w_2 s_{k-1}}_{\text{as in AR(2)}} + \underbrace{w_3 s_k^2 + w_4 s_k s_{k-1} + w_5 s_{k-1}^2}_{\text{new: products}}$



$L = 3$: each input feeds its linear feature (blue) and the products that contain it (orange); only the readout weights are trained

- with the products the forecast can follow a curved cloud of points instead of a flat plane
- $1 + L + L(L+1)/2$ features: 6 for $L = 2$, 153 for $L = 16$

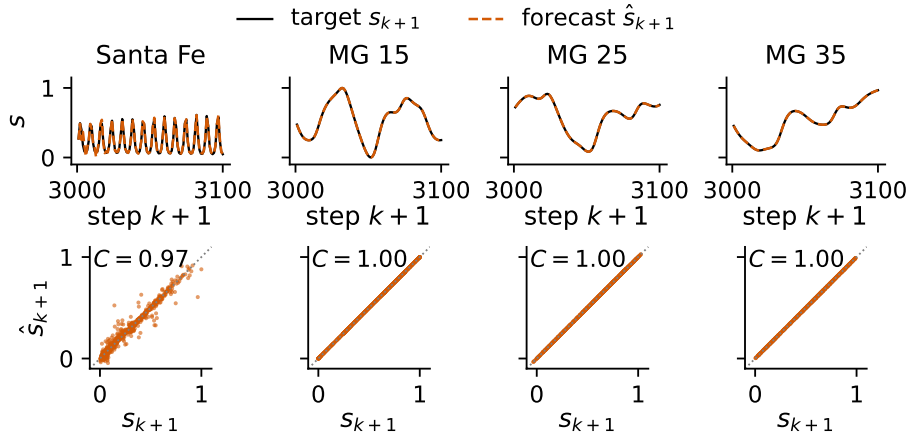
Loss: ridge regression

$$\mathcal{L}(\mathbf{w}_\tau) = \sum_k \left(s_{k+\tau} - \mathbf{w}_\tau^\top \varphi_k\right)^2 + \lambda \|\mathbf{w}_\tau\|^2$$

$\lambda > 0$ keeps the weights small; solved by one linear system

A. E. Hoerl, R. W. Kennard, Technometrics 12, 55 (1970)

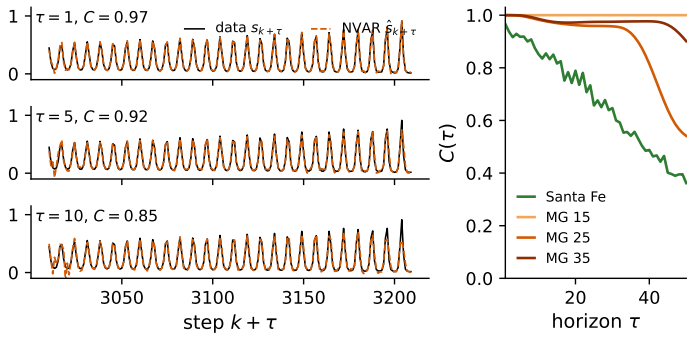
One-Step NVAR Forecasts



L, stride and λ chosen on the validation block. On Santa Fe the points stay near the diagonal also at the high flashes, where the linear models bend away.

NVAR Forecast τ Steps Ahead

$\hat{s}_{k+\tau} = \mathbf{w}_\tau^\top \boldsymbol{\varphi}_k$ one ridge weight vector per horizon τ



Santa Fe: $C = 0.92$ at $\tau = 5$, 0.85 at $\tau = 10$ (AR: 0.60); Mackey–Glass: $C \geq 0.99$ at $\tau = 5$

6

Neural Networks

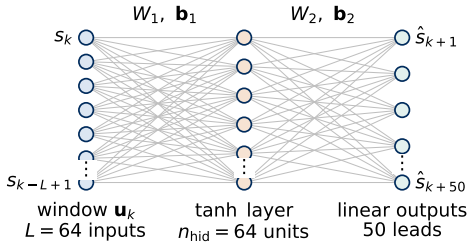
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Neural Networks for Forecasting

Every forecast so far is a linear readout of fixed features of the past. Neural networks **learn** the features.

1. **MLP**, multilayer perceptron: learned features of a window of past values
2. **RNN**, recurrent neural network: a hidden state that carries the past
3. **LSTM**, long short-term memory: a recurrent network with gates and a long memory
4. **neural ODE**: a network for the law of motion **F**
5. **PINN**, physics-informed neural network: a network for the solution, trained with the law

MLP Forecaster



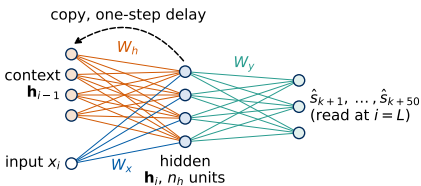
Multilayer perceptron (MLP)

$$\hat{\mathbf{y}}_k = W_2 \tanh\left(W_1 \mathbf{u}_k + \mathbf{b}_1\right) + \mathbf{b}_2$$

$\mathbf{u}_k = (s_k, s_{k-1}, \ldots, s_{k-L+1})$: window of the last L values; $\hat{\mathbf{y}}_k = (\hat{s}_{k+1}, \ldots, \hat{s}_{k+50})$: one output per horizon τ

- NVAR with learned features $\tanh(W_1 \mathbf{u}_k + \mathbf{b}_1)$
- $L = n_{\text{hid}} = 64$: 7410 parameters, gradient descent on the error of all 50 horizons
- memory limited to the window

Recurrent Cell



Recurrent neural network (RNN)

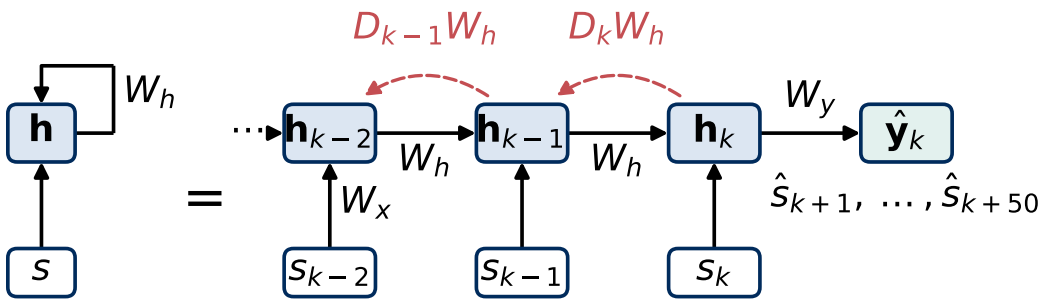
$$\mathbf{h}_k = \tanh\left(W_h \mathbf{h}_{k-1} + W_x s_k + \mathbf{b}\right), \quad \hat{\mathbf{y}}_k = W_y \mathbf{h}_k + \mathbf{b}_y$$

$W_h \in \mathbb{R}^{n_h \times n_h}$: feedback; $W_x \in \mathbb{R}^{n_h}$: input; $W_y, \mathbf{b}_y$: readout of the 50 horizons

- the state $\mathbf{h}_k$ is fed back at the next step, so it depends on $s_k, s_{k-1}, \ldots$
- the parameter count does not grow with the history; $\mathbf{h}_k \in (-1, 1)^{n_h}$

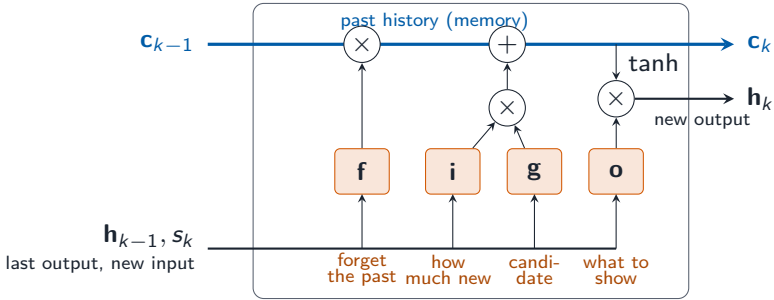
J. L. Elman, Cogn. Sci. 14, 179 (1990), doi:10.1207/s15516709cog1402_1

Unrolling a Recurrent Network



The same cell and weights act at every step. The final hidden state supplies the forecast vector. Dashed red arrows mark the derivative factors used in training through time, with $D_k = \operatorname{diag}\left(1 - \tanh^2(W_h \mathbf{h}_{k-1} + W_x s_k + \mathbf{b})\right)$, so that $\partial \mathbf{h}_k / \partial \mathbf{h}_{k-1} = D_k W_h$.

LSTM Cell: Gates and Memory



- **longer memory** of past inputs and states: $\mathbf{c}$ is kept while the forget gate stays near 1
- the additive update of $\mathbf{c}$ lets gradients pass many steps without vanishing (explosion: gradient clipping)

S. Hochreiter, J. Schmidhuber, Neural Comput. 9, 1735 (1997)

LSTM Cell: Equations

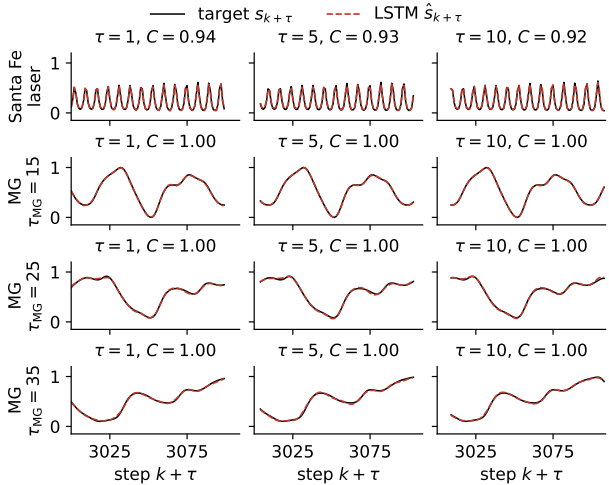
One step of the cell, input $[\mathbf{h}_{k-1}, s_k]$: last output and new value

$$\mathbf{f}_k = \sigma(W_f[\mathbf{h}_{k-1}, s_k] + \mathbf{b}_f) \quad \text{keep / forget the past}$$
$$\mathbf{i}_k = \sigma(W_i[\mathbf{h}_{k-1}, s_k] + \mathbf{b}_i) \quad \text{how much new to write}$$
$$\mathbf{g}_k = \tanh(W_g[\mathbf{h}_{k-1}, s_k] + \mathbf{b}_g) \quad \text{candidate new content}$$
$$\mathbf{o}_k = \sigma(W_o[\mathbf{h}_{k-1}, s_k] + \mathbf{b}_o) \quad \text{how much memory to show}$$
$$\mathbf{c}_k = \mathbf{f}_k \odot \mathbf{c}_{k-1} + \mathbf{i}_k \odot \mathbf{g}_k \quad \text{past kept + new written}$$
$$\mathbf{h}_k = \mathbf{o}_k \odot \tanh(\mathbf{c}_k) \quad \text{new output}$$

σ: sigmoid, values in $(0, 1)$; $\odot$: product component by component; W, $\mathbf{b}$: trainable weights

- forget gate near 1 and input gate near 0: the memory $\mathbf{c}$ is carried unchanged
- the readout of the last $\mathbf{h}_k$ gives the forecasts $\hat{s}_{k+\tau}$, as for the RNN

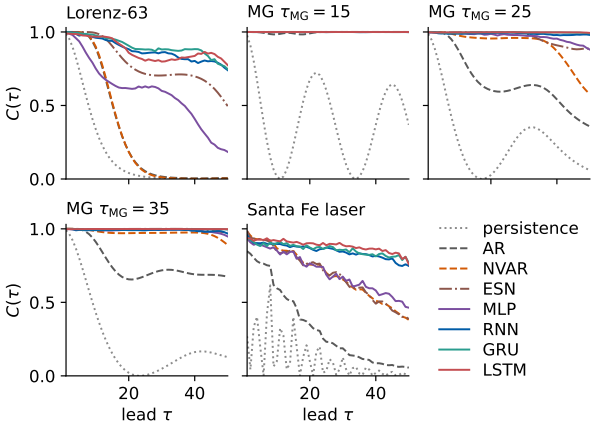
LSTM Forecasts on Four Signals



rows: Santa Fe and MG at $\tau_{\mathrm{MG}} = 15, 25, 35$; columns: horizons $\tau = 1, 5, 10$; black: target $s_{k+\tau}$, dashed: forecast $\hat{s}_{k+\tau}$, first 100 test steps; C on the whole test block

- MG: $C \geq 0.99$ up to $\tau = 10$ in all three regimes
- Santa Fe: C from 0.94 to 0.92, the flattest decay of the three networks

LSTM Capacity on All Signals



$C(\tau)$ on the test block, LSTM in red

C_{tot} of the LSTM (max 50):

- Lorenz-63: 43.8 (AR 15.4)
- MG 15, 25, 35: 50.0, 49.9, 49.9
- Santa Fe: 44.3 (AR 16.5)

ESN and GRU: further recurrent models, not discussed here

7

Neural ODEs and Physics-Informed Networks

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Physics Knowledge in the Model

model	known	learned
AR, MA, NVAR	only the record s_k	linear readout of fixed features of $\mathbf{u}_k$
MLP, RNN, LSTM	only the record s_k	features and readout
Neural ODE	form $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$	vector field $\mathbf{F}_\theta$
PINN	equation, initial state	solution $x_\theta(t)$

- first: models that see only the record, from fixed to learned features
- then: models with physics inside, from the full equation to a missing term
- each method is compared with a standard method for the same task

From the Record to the Laws of Motion

So far: the record only

- models learn the forecast map G_τ from $s_1, \ldots, s_k$, nothing else
- they need long records, know nothing beyond the data, and chaos limits their horizon

Often more is known

- the law of motion, e.g. Newton's equation, a diffusion equation, a conservation law
- or part of it: the form of the equation, some terms, some parameters

- next: put the known physics into the learning: the form of the equation (Neural ODE), the full equation (PINN)

Neural ODE

Recall: equations of motion

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = \mathbf{F}(\mathbf{x})$$

$\mathbf{F}$: velocity of the state, often unknown

Model $\mathbf{F}$ by a neural network

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = \mathbf{F}_\theta(\mathbf{x})$$

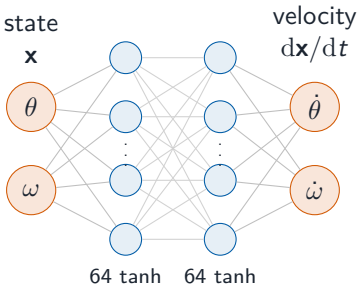
input the state, output its velocity

Training: integrate, compare with data

$$\hat{\mathbf{x}}(t_k) = \text{ODE solver}(\mathbf{F}_\theta, \mathbf{x}_0, t_k), \qquad \mathcal{L}(\theta) = \tfrac{1}{K}\sum_{k=1}^{K} \left\|\hat{\mathbf{x}}(t_k) - \mathbf{y}_k\right\|^2$$

$\mathbf{y}_k$: measured states; gradients flow back through the solver

Neural ODE: a Network for $\mathrm{d}\mathbf{x}/\mathrm{d}t$



pendulum: $\mathbf{x} = (\theta, \omega)$, 4482 weights

Unknown $\mathbf{F}$ modelled as a neural network

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = \mathbf{F}(\mathbf{x}) \;\approx\; \mathbf{F}_\theta(\mathbf{x})$$

input: the state; output: its velocity

Time evolution: forward Euler

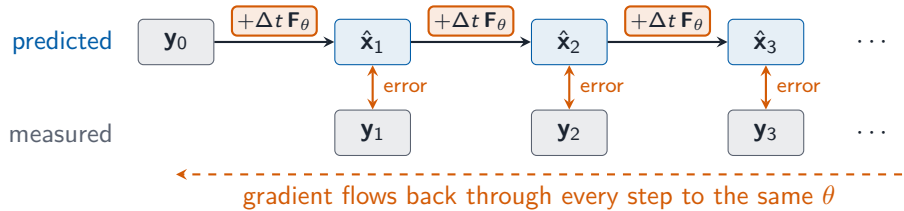
$$\mathbf{x}_{k+1} = \mathbf{x}_k + \Delta t\, \mathbf{F}_\theta(\mathbf{x}_k)$$

notebook: RK4, same idea

Discovery

after training, $\mathbf{F}_\theta$ is the vector field of the system, extracted from data

Training a Neural ODE



Loss

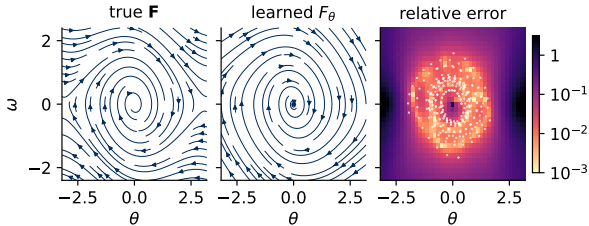
$$\mathcal{L}(\theta) = \sum_{k=1}^{n} \left\| \hat{\mathbf{x}}_k - \mathbf{y}_k \right\|^2$$

1. start from a measured state $\mathbf{y}_0$ and roll out $n = 10$ steps with the current θ
2. compare every $\hat{\mathbf{x}}_k$ with the measured $\mathbf{y}_k$: the loss
3. backpropagate through all steps, update θ once; repeat on many segments

Learning the Damped Pendulum

The unknown vector field, discovered from data

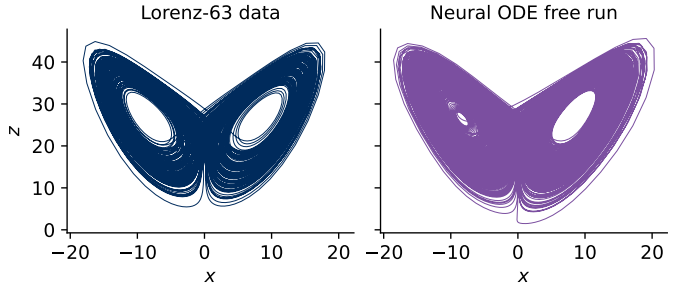
from 8 noisy trajectories alone the network recovers the field **F** of the pendulum



state $(\theta, \omega = \dot{\theta})$; 8 noisy trajectories; left: true field, middle: learned $\mathbf{F}_\theta$, right: $|\mathbf{F}_\theta - \mathbf{F}|/|\mathbf{F}|$, dots: training data

- median relative error 0.012 near the data, 0.23 away from it
- the field is learned where the data are; near the saddles at $\theta = \pm\pi$ it is a guess

Learned Lorenz Attractor



- a continuous field can be analysed: fixed points by Newton's method, Jacobians, Lyapunov exponents
- not exact: the learned C_+ is a stable spiral, the true one weakly unstable, and the left lobe gets an extra dense spiral

Reading the Learned Field $\mathbf{F}_\theta$

The network cannot be read directly, but every physical quantity of a vector field can be computed from it.

1. **look at it**: learned against true field and the error map; trust it near the data only
2. **invariants of the dynamics** from $\mathbf{F}_\theta$, Lorenz-63: largest Lyapunov exponent 0.899 ± 0.010 (true 0.906 ± 0.007); volume contraction $\nabla \cdot \mathbf{F}_\theta = -13.88$ (true $-41/3 = -13.67$)
3. **fixed points and stability**: solve $\mathbf{F}_\theta(\mathbf{x}^*) = 0$, eigenvalues of the Jacobian $\partial \mathbf{F}_\theta / \partial \mathbf{x}$ by automatic differentiation
4. **a formula**: sparse regression of $\mathbf{F}_\theta$ on a library of candidate terms $\{1, x, x^2, \sin x, \dots\}$ keeps only a few terms

   hybrid model, known pendulum plus a network for the unknown friction: one term survives, $-c\,\dot\theta|\dot\theta|$ with $c = 0.1497$ (true 0.15)

S. L. Brunton, J. L. Proctor, J. N. Kutz, PNAS 113, 3932 (2016)

Known Laws, Few Measurements

system	known law	measured	wanted
oscillator	$m\ddot{x} + c\dot{x} + kx = 0$	noisy positions	$x(t)$, damping c
heat in a rod	$\partial_t u = D\partial_x^2 u$	thermometers	$u(x,t)$, diffusivity D
fluid flow	Navier–Stokes	dye concentration	velocity and pressure

- forward: solve the known equation from initial/boundary conditions
- inverse: infer an unknown parameter from observations
- data assimilation: combine sparse observations with the law

M. Raissi *et al.*, Science 367, 1026 (2020); G. E. Karniadakis *et al.*, Nat. Rev. Phys. 3, 422 (2021)

Residual of the Law

Insert the candidate curve into the equation

$$r_\theta(t) = \frac{m\ddot{x}_\theta(t) + c\dot{x}_\theta(t) + kx_\theta(t)}{k}$$

m: mass; c: damping; k: spring constant. Dividing by k gives the residual units of position.

- evaluate the network and its time derivatives at chosen collocation times
- the equation supplies a target residual of zero; measured positions are unnecessary there
- test unused times and initial conditions as well as the training residual

Two curves for $\ddot{x} + 0.4\dot{x} + 4x = 0$, $x(0) = 1$, $\dot{x}(0) = 0$

$x(t) = 0$: residual zero, but wrong initial position.
$x(t) = \cos(2t)$: correct initial conditions, but $r(t) = -0.2\sin(2t)$.
The desired motion must satisfy both tests.

Derivatives by Automatic Differentiation

One hidden layer, written out

$$x_\theta(t) = \sum_j a_j \tanh(w_j t + b_j) + c_0$$

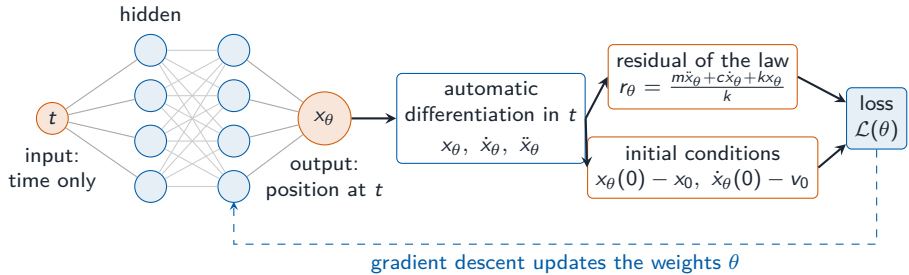
$$\dot{x}_\theta(t) = \sum_j a_j w_j (1 - \tanh^2(w_j t + b_j))$$

$$\ddot{x}_\theta(t) = -2 \sum_j a_j w_j^2 \tanh(w_j t + b_j)(1 - \tanh^2(w_j t + b_j))$$

$\theta = \{a_j, w_j, b_j, c_0\}$; the same chain rule applies through any number of layers

- derivatives in t: evaluate velocity and acceleration by automatic differentiation, up to floating-point error
- gradients in θ: differentiate the loss to update the network weights

Physics-Informed Neural Network (PINN)



PINN

$$\mathcal{L}(\theta) = \frac{1}{N_r}\sum_{i=1}^{N_r} r_\theta(t_i)^2 + \lambda\Big([x_\theta(0) - x_0]^2 + \frac{[\dot{x}_\theta(0) - v_0]^2}{\omega_0^2}\Big)$$

t_i: chosen collocation times; λ: initial-condition weight; $\omega_0 = \sqrt{k/m}$

Lagaris *et al.*, IEEE TNN 9, 987 (1998); Raissi *et al.*, J. Comput. Phys. 378, 686 (2019)

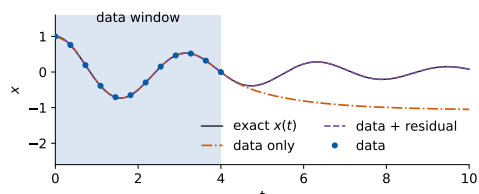
Data Plus the Law

The law as a soft constraint in the loss

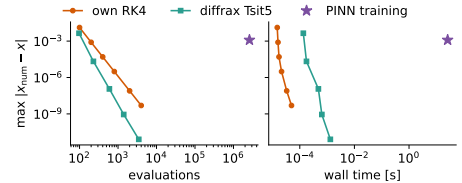
$$\mathcal{L}(\theta) = \underbrace{\frac{1}{12}\sum_{i=1}^{12}\left(x_\theta(t_i) - x(t_i)\right)^2}_{\text{data}} + \underbrace{\frac{1}{N_r}\sum_{j=1}^{N_r} r_\theta(t_j)^2}_{\text{residual of the law}}$$

data: the 12 pairs $(t_i, x(t_i))$ on $[0, 4]$; collocation times t_j: N_r points on $[0, 10]$, residual only, no target

- same samples, same network; only the residual term is added
- the law is imposed through a penalty
- beyond the data: error 1.2×10^{0} with data only, 3.0×10^{-4} with the law



PINN: Forward and Inverse Problems



forward: known c, solve for $x(t)$
RK4, $h = 0.1$: error 5.0×10^{-5} with 400 evaluations; PINN: 1.2×10^{-3} after about 2.6×10^{6}
a classical solver wins by far

inverse: unknown c, trained next to θ from noisy data
PINN $c = 0.401 \pm 0.006$;
least-squares fit of the exact solution 0.400 ± 0.004; true 0.4
the PINN finds an unknown parameter of a known law

Strengths and Limits of PINNs

Strengths

- estimate unknown coefficients of a known law
- combine sparse, noisy data with the residual
- evaluate a continuous field at arbitrary coordinates

Limits

- forward solves are much slower in these tests
- loss terms can produce conflicting gradients
- small mean residual can accompany wrong motion

S. Wang *et al.*, SIAM J. Sci. Comput. 43, A3055 (2021); G. E. Karniadakis *et al.*, Nat. Rev. Phys. 3, 422 (2021)

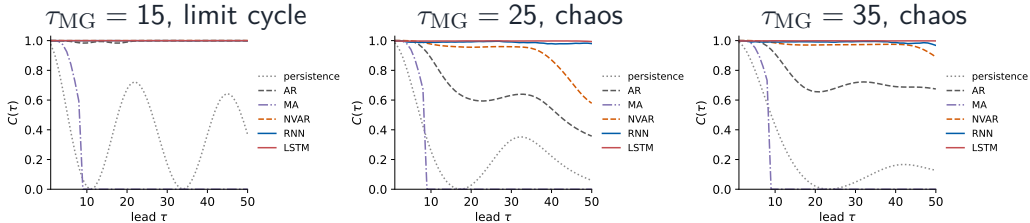
8

Comparison on Lorenz and Mackey–Glass



JAGIELLONIAN UNIVERSITY
IN KRAKOW

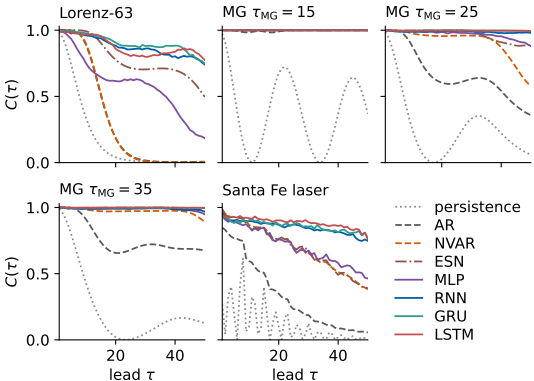
Mackey–Glass: Periodic and Chaotic



capacity $C(\tau)$ against the lead τ on the test block

- limit cycle: AR, NVAR and both networks stay near $C = 1$ at all leads; MA forgets after its 8 steps
- chaos: AR decays ($C_{\mathrm{tot}} = 33.1$ at 25), NVAR holds longer (45.5), RNN and LSTM stay near 1 (49.4, 49.9)
- the same ordering at 35 (AR 38.0, NVAR 48.6, LSTM 49.9)

Capacity of All Models



Direct forecasts on common test origins

$C_{\text{tot}} = \sum_{\tau=1}^{50} C(\tau)$, maximum 50

model	MG 15	MG 25	MG 35
persistence	19.5	13.3	11.8
AR	49.8	33.1	38.0
MA	7.1	7.2	7.4
NVAR	50.0	45.5	48.6
RNN	50.0	49.4	49.5
LSTM	50.0	49.9	49.9

- limit cycle: every fitted model except MA near the maximum
- chaos: RNN and LSTM, models with a state, gain most

Summary

- A time series measures a projection of a hidden state: $s_k = h(\mathbf{x}(k\Delta t)) + \eta_k$.
- We forecast a future value from past observations. Every model is trained and tested in time order and scored by the capacity $C(\tau)$.
- AR uses past values and MA uses past forecast errors. ARMA combines the two, and NVAR adds nonlinear features with a linear readout.
- MLPs learn window features. RNNs and LSTMs build a hidden state while reading the window and are trained by backpropagation through time.
- Neural ODEs learn a vector field. PINNs learn a solution constrained by a given equation and conditions or data.
- In chaos, initial uncertainty grows. Finite precision limits the useful forecast horizon even with the correct law.

Thank You